Probability Gymnastics

This is a selection of exercises on probability theory that will allow you to find out how fluent you are in manipulating probabilities. Except for the first one, the exercises are all taken from the textbook *Probability and Statistics* by De Groot and Schervish. The number at the beginning of each item refers to the section in the textbook where you could read up on this type of problems if necessary.

1. Explain the following concepts:
   a) Sample space b) Event c) Random variable
2. (1.5) One ball is to be selected from a box containing red, white, blue, yellow and green balls. If the probability that the selected ball will be red is $1/5$ and the probability that it will be white is $2/5$, what is the probability that it will be blue, yellow, or green?
3. (1.6) If two balanced dice are rolled, what is the probability that the sum of the two numbers that appear will be even?
4. (1.6) If two balanced dice are rolled, what is the probability that the difference of the two numbers that appear is less than 3?
5. (2.1) If $A \subset B$ with $\Pr(B) > 0$, what is the value of $\Pr(A|B)$?
6. (2.1) If A and B are disjoint events and Pr(B) > 0, what is the value of Pr(A|B)?
7. (2.1) A box contains three coins with a head on each side, four coins with a tail on each side, and two fair coins. If one of these nine coins is selected at random and tossed once, what is the probability that a head will be obtained?
8. (2.1) Suppose that a box contains one blue card and four red cards, which are labelled A, B, C, and D. Suppose also that two of these five cards are selected at random, without replacement.
   a) If it is known that card A has been selected, what is the probability that both cards are red?
   b) If it is known that at least one red card has been selected, what is the probability that both cards are red?
9. (2.2) If three balanced dice are rolled, what is the probability that all three numbers will be the same?
10. (2.3) A new test has been devised for detecting a particular type of cancer. If the test is applied to a person who has this type of cancer, the probability that the person will have a positive reaction is 0.95 and the probability that the person will have a negative reaction is 0.05. If the test is applied to a person who does not have this type of cancer, the probability that the person will have a positive reaction is 0.05 and the probability that the person will have a negative reaction is 0.95. Suppose that in the

general population, one person out of every 100,000 people has this type of cancer. If a person selected at random has a positive reaction to the test, what is the probability that they have this type of cancer?

11. (2.3) Suppose that 30 percent of the bottles produced in a certain plant are defective. If a bottle is defective, the probability is 0.9 that an inspector will notice it and remove it from the filling line. If a bottle is not defective, the probability is 0.2 that the inspector will think that it is defective and remove it from the filling line.

    a) If a bottle is removed from the filling line, what is the probability that it is defective?

    b) If a customer buys a bottle that has not been removed from the filling line, what is the probability that it is defective?